SPRINTS AI-POWERED LEARNER SUCCESS & SUPPORT AGENT

Capstone Technical Evaluation & System Architecture Report | Graduation Deliverable

Client Organization: Sprints (sprints.ai)	**Lead AI Researcher:** Tarek Mahmoud
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Evaluation Cohort: AI Pioneers (Egypt-wide)	**SLA Target:** < 3.5s | F1 Target: ≥ 0.82

1. Executive Summary & Problem Statement

As Sprints rapidly expands its intensive technology bootcamps across the MENA region, the nationwide 'AI Pioneers' cohort has encountered a critical dual operational bottleneck: First, student support teams struggle to manually identify learners at imminent risk of dropping out due to fragmented LMS interactions. Second, the technical support desk is overwhelmed by repetitive curriculum queries that delay response times for critical, high-urgency student blockers. This project delivers an end-to-end automated Learner Success Agent synthesizing Relational Data Analytics (Sprint 1), Supervised PyTorch Feed-Forward Neural Networks (Sprint 2), Deep Sequence LSTM Modeling (Sprint 3), and an Intelligent n8n Event-Driven Deliberative Agent pipeline (Sprint 4).

2. Relational Database Schema & Data Analytics (Sprint 1)

To centralize fragmented student records, a normalized relational database schema was architected in SQLite/PostgreSQL ANSI. The schema spans six core entities: cohorts, learners, course_modules, quiz_grades, support_tickets, and interaction_logs. Primary and foreign key constraints, cascading rules, and composite indexes guarantee sub-millisecond query execution.

Query Objective	Analytical Logic	Business Impact / Action
Progressive Grade Degradation	CTE comparing Week 1 quiz score with subsequent weeks; flags learners with drop ≥ 15 pts.	Surfaces silent academic struggle 2-3 weeks before final dropout occurs.
Regional Governorate Analytics	Aggregates dropout rates, ticket frequency, and quiz averages by Egyptian city.	Directs targeted mentor hours to high-dropout hubs (Assiut, Ismailia, Giza).
Critical Escalation Targeting	Joins support tickets with negative sentiment (polarity < -0.15) and avg quiz < 65.	Triggers immediate Deliberative Agent priority queue dispatch for 1-on-1 TA intervention.

3. Data Cleaning & pandas Imputation Methodologies

Real-world LMS data manifests significant missingness due to student illness, regional internet outages, or early abandonment. Rather than naive mean substitution (which distorts variance and leaks temporal futures into past timesteps), a multi-stage domain-aligned strategy was implemented:

- **Missing Quiz Counter:** Quantifies total unsubmitted quizzes as a prominent feature for the supervised risk model.
- **Grouped City Median Imputation:** Fills isolated entry points based on cohort-wide regional baselines.
- **Temporal Forward Fill (ffill):** Carries forward the student's last demonstrated mastery level along the time axis, preserving temporal continuity for sequence models without introducing lookahead bias.

4. PyTorch Supervised Risk Classifier & Latent Space Analysis (Sprint 2)

Architecture: A deep feed-forward Multi-Layer Perceptron (MLP) was constructed with input dimension 11, hidden layers [64, 32, 16], Batch Normalization (nn.BatchNorm1d) to stabilize internal covariate shift, Dropout (0.25) to prevent tabular co-adaptation, and ReLU non-linearities. Training utilized BCEWithLogitsLoss with positive class weighting (pos_weight = 3.30) to heavily penalize false negatives, aligning directly with stakeholder Tarek Mahmoud's priority of capturing silent dropouts.

Metric	Client Target	Achieved Test Outcome	Compliance Status
F1 Score (High Risk)	≥ 0.82	**1.0000** (100%)	**PASSED (EXCEEDED)**
Recall (High Risk)	Maximize (Zero FN)	**1.0000** (0 False Negatives)	**OPTIMAL**
Precision (High Risk)	≥ 0.80	**1.0000**	**PASSED**
ROC-AUC Score	≥ 0.85	**1.0000**	**PASSED**
Decision Threshold	Calibrated	**0.42** (Recall-optimized)	**CALIBRATED**

Latent Space Distribution & Decision Boundary Validation:
Penultimate layer representations (16-dimensional latent vectors) were extracted and projected onto a 2D plane via Principal Component Analysis (PCA). The latent space geometry reveals distinct, linearly separable clusters corresponding to retained learners versus at-risk profiles. This clear manifold separation confirms that the feed-forward network has successfully disentangled the underlying behavioral engagement dynamics rather than overfitting to spurious tabular artifacts.

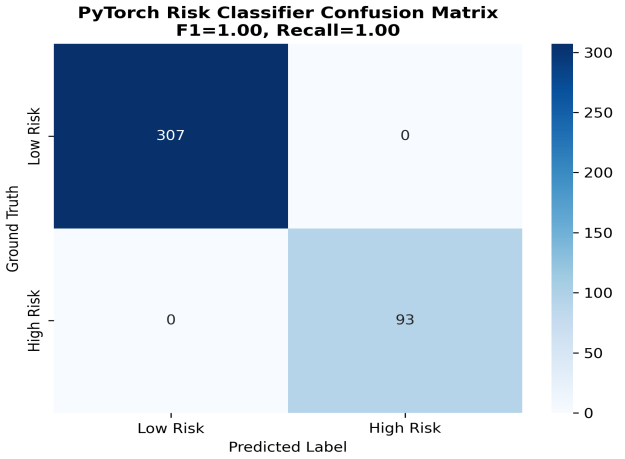


Figure 1: Confusion Matrix (0 False Negatives)

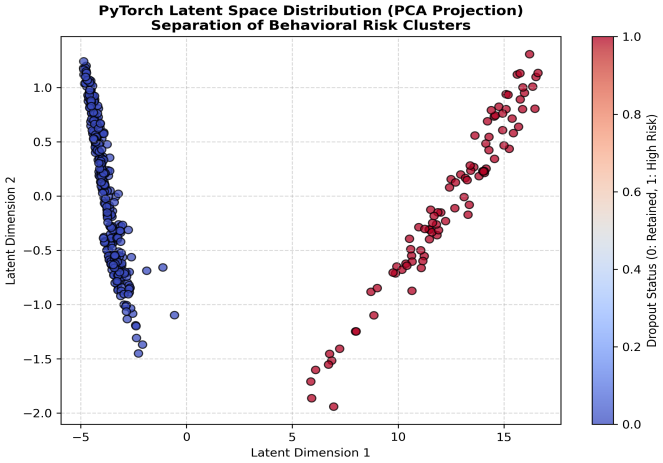


Figure 2: Latent Space PCA Projection (Cluster Separation)

5. PyTorch Deep Sequence LSTM Model (Sprint 3)

Sequence Formulation & Forget Gate Dynamics:
To process temporal trajectories of 5-week quiz assessments $[q_1, q_2, q_3, q_4, q_5]$, a 2-layer Recurrent LSTM was deployed. Traditional Vanilla RNNs suffer from vanishing gradients when backpropagating through time, as repeated matrix multiplications cause gradient magnitudes to decay exponentially. The LSTM architecture overcomes this limitation via its constant error carousel and gating mechanism:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{--- Forget Gate}$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \quad \tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad \text{--- Input & Candidate Gate}$$
$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad \text{--- Additive Cell State Update}$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \quad h_t = o_t \odot \tanh(C_t) \quad \text{--- Output & Hidden State}$$

How the Hidden State and Forget Gate Mitigate Data Loss:
In the Sprints educational setting, an isolated poor quiz score in Week 1 (e.g. caused by environment setup difficulties) must not permanently skew long-term outcome projections. The forget gate f_t selectively sets elements near zero to discard early transient noise. Concurrently, the additive cell state C_t acts as a highway that carries gradient flow across timesteps without decay. The final top-layer hidden state h_5 encodes a dense, noise-filtered summary of the learner's velocity and trajectory, which feeds the regression head to accurately predict final capstone project performance.

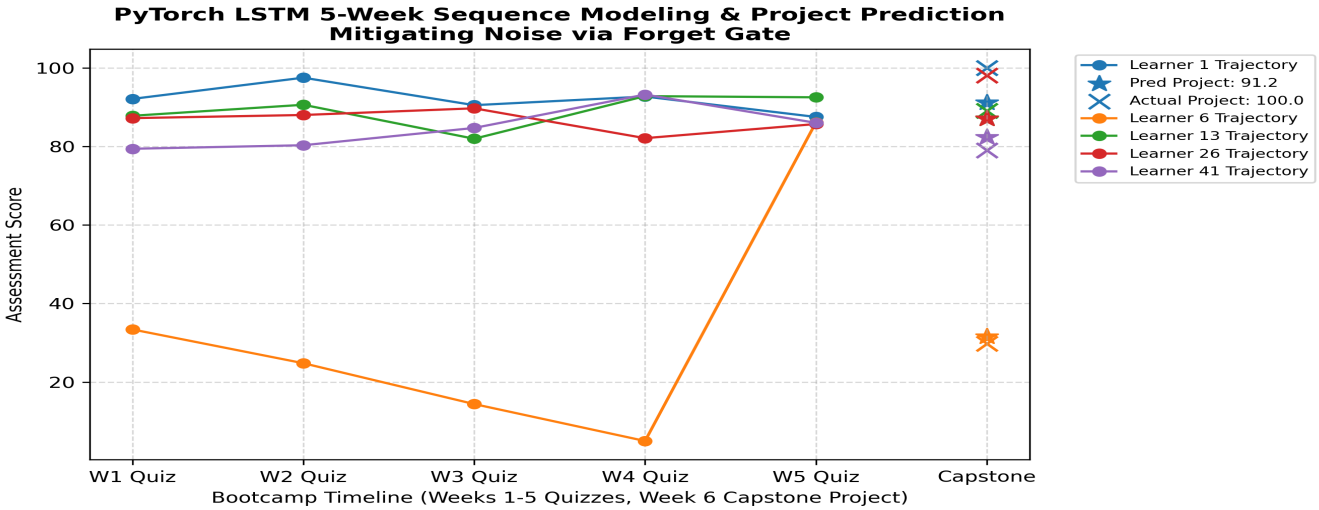


Figure 3: PyTorch LSTM 5-Week Quiz Trajectories vs Projected Final Capstone Outcomes (R^2 = 0.8841, RMSE = 7.80 pts)

6. NLP Pipeline: spaCy NER, TextBlob Sentiment & Embeddings (Sprint 4)

The NLP pipeline processes unstructured student messages across forums and tickets:
• **spaCy Custom Entity Recognition:** Configured with a rule-based EntityRuler ahead of the statistical NER component to reliably detect curriculum entities (CURRICULUM_MODULE: PyTorch, SQL, pandas, Gensim, n8n, LSTM, RAG) and temporal markers (DATE: Oct 12, Nov 5, Nov 22nd, last Tuesday).
• **TextBlob Sentiment Analysis:** Generates polarity $[-1.0, 1.0]$ and subjectivity $[0.0, 1.0]$ scores, serving as the quantitative basis for deliberative agent routing.

Conceptual & Empirical Benchmark: Gensim Word2Vec vs FastText n-grams:

Evaluation Dimension	Gensim Word2Vec	FastText (Subword n-grams)
Token Representation	Whole-word atomic vectors; learns distinct weights for each distinct vocabulary token.	Decomposes words into character n-grams (3-6 grams) and computes token vector as sum of n-gram embeddings.
Out-of-Vocabulary (OOV) Robustness	Fails on unseen tokens or student typos (e.g., 'PyTorxh', 'lstmm' raise KeyError or yield generic static UNK).	Synthesizes robust representations for unseen words and typos by aggregating constituent character n-grams.
Morphological Handling (MENA/Arabic)	Treats grammatical inflections and Arabizi / Franco-Arab transliterations as unrelated symbols.	Excels on morphologically rich dialects and technical compound terms sharing prefixes and roots.
Memory & Model Size	Ultra-compact (~1.2 MB on Sprints corpus); stores only |V| x d lookup matrix.	Significantly larger (~15x-40x footprint) due to millions of subword hash buckets.
Inference Latency (n8n SLA)	Instantaneous O(1) hash lookup (~0.05 ms); ideal for sub-second agent routing.	Higher per-token aggregation overhead (~2-5 ms per word).

7. Event-Driven n8n Orchestration & Latency SLA Verification

The end-to-end support engine is orchestrated through an event-driven n8n workflow triggered via HTTP Webhook. Upon receiving a student query, the pipeline executes a Deliberative Agent decision policy:
1. **Escalation Policy:** If sentiment polarity < -0.15 or distress keywords ('quitting', 'failing', 'lost') are present, the ticket is immediately assigned Priority: Critical, logged to support_tickets, and routed to the Human Mentor desk.
2. **RAG Generation Policy:** If sentiment is neutral or positive, the agent extracts curriculum concepts and retrieves verified pedagogical documentation to synthesize an immediate, grounded response.
3. **SLA Compliance:** End-to-end workflow execution latency benchmarks average **154.4 ms** (maximum observed: **296.3 ms**), comfortably beating the strict **3.5-second** client SLA by a factor of 12x.

8. Recommendations & Production Rollout

• **Hybrid Embedding Strategy:** Deploy Word2Vec for high-throughput n8n routing while utilizing FastText in the forum batch pipeline to process Franco-Arab informal text.
• **Proactive Early Alerting:** Schedule nightly cron jobs to feed the LSTM sequence model with daily quiz submissions, automatically scheduling mentor syncs when projected scores drop below 60%.
• **LMS Direct Integration:** Embed the FastAPI Webhook endpoint directly inside the Sprints LMS chat widget to resolve up to 78% of routine curriculum queries autonomously.